Abhigyan Gandhi

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PROFESSIONAL SUMMARY

Software engineer specialising in high-performance C++ and systems-level programming, with strong fundamentals in algorithms, data structures, and low-level optimization. MSc from the University of Leeds and recipient of the Game Republic Red Kite Games Technology Award 2025. Experienced across C++, C#, Python, and CUDA, and comfortable working close to the hardware.

SKILLS

- **Languages:** C++ (modern, C++17/20), C#, Python, CUDA, GLSL, HLSL, TypeScript, SQL, Bash.
- **Systems & Performance:** Multithreading, SIMD, memory management, cache-aware design, parallel computing, low-level optimization, profiling and debugging.
- **Graphics & GPU:** Vulkan, DirectX 12, OpenGL, SPIR-V, OpenCL, ray tracing, PBR, render graphs, GPU synchronization and descriptors, shader programming.
- **Engines & Frameworks:** Unreal Engine, Unity, ECS architecture, physics simulation, networking.
- **Tools:** Git, Linux, Docker, GitLab CI/CD, Azure DevOps, RenderDoc, Nsight, Tracy, Node.js, MATLAB, MySQL.
- **AI/ML:** TensorFlow, Scikit-Learn, Pandas, CNNs, GANs, computer vision.

PROJECTS

- **Game Engine Development (C++, Vulkan, 2025):** Developed multi-pass Vulkan renderer with PBR pipeline; implemented GPU culling, ECS and vehicular physics simulation; achieved **100+ FPS** on RTX hardware; won **Game Republic Red Kite Games Game Technology Award** at the **Student Showcase 2025** (Group project). [GitHub Repo]
- **Vulkan Ray Tracer (C++, Vulkan, 2026):** Built a hardware-accelerated ray tracer using **VK_KHR_ray_tracing_pipeline**; implemented BLAS/TLAS, Shader Binding Table, and **recursive ray tracing (shadows, reflections)**. [GitHub Repo]
- **LETTER SHIFT (Unity, C#, HTML, 2025):** Developing a daily puzzle game for mobile and web; preparing for release on **letter-shift.com** and Android (APK available on portfolio). [GitHub Repo]
- **Ray Tracing and Rasterization (C++, OpenGL, Vulkan, 2024):** Implemented CPU-based ray tracer and GPU rasterizer to compare performance and visual fidelity. Reduced GPU render pass time by **35%** through descriptor sets.
- **Bezier Curves and Physics Simulation (C++, OpenGL, 2024):** Created a Bezier curve renderer and physics-based animation with collision detection for a character and bouncing dodecahedra.

WORK EXPERIENCE

Tech Designer (Freelance) | **Creative AF Games** | Mar 2026 - Present | Leeds, UK (Remote)

- Engineered character ability systems and gameplay mechanics in **Unreal Engine 5** using **Blueprints** and **C++**.
- Optimised runtime performance identifying and resolving **CPU/GPU bottlenecks** to maintain frame budget targets.
- Collaborated with design and engineering teams in an **agile pipeline** to implement and iterate on technical features.

Ticket Sales Advisor (Part-time) | **Leeds United** | Oct 2025 - Present | Leeds, UK

- Resolved **time sensitive problems efficiently** in a fast-paced, high pressure environment, ensuring a positive exp.
- Delivered **calm, professional customer service**, assisting supporters with ticketing issues, collections and enquiries.

Game Development Intern | **Sentient Labs** | Jun 2020 - Aug 2020 | Dubai, UAE

- Developed a **WebGL** game in **Unity**; lifted user engagement by **25%** and boosted average session duration by **18%**.
- Implemented responsive gameplay mechanics and modular UI systems, reducing UI load time by **40%**.
- Integrated **50+ art and audio assets**, ensuring visual consistency at a sustained **60 FPS**.
- Collaborated in agile sprints, completing **100%** of assigned tasks using Git-based version control.

AI/ML Intern | **Stella Stays** | Aug 2021 - Jan 2022 | Dubai, UAE

- Implemented predictive pricing model using **LSTM** and **regression,** improving base price accuracy to **10% MAPE**.
- Built a NLTK-based sentiment analysis pipeline for hotel reviews, achieving **80% accuracy**.
- Developed a room allocation algorithm based on **minimum slack** and **clustering**; led to a **+10% optimized** allocator.

EDUCATION

MSc. High-Performance Graphics and Games Engineering, Merit | University of Leeds | Sept 2024 - Sept 2025

- **Modules**: Foundation of Computer Graphics, Advanced Rendering, Modelling and Animation.
- **Dissertation**: Evaluating the performance and visual fidelity of INTEL OIDN against NVIDIA NRD. (Key finding: Hybrid system including real-time and offline denoising is recommended for best performance)

B.E. (Hons) Computer Science, Data Science, CGPA: 8.72 | BITS PILANI DUBAI CAMPUS | Sept 2018 - Sept 2022

- **Modules**: Machine Learning, Data Mining, Foundations of Data Science, Neural Networks.
- **Final Year Design Project**: Mobile Robot Path Planning using Deep Learning Techniques.